

CENTENNIAL JING-ZHANG AI INNOVATION BELT

JING-ZHANG OPEN GROUND

JING-ZHANG OPEN GROUND

Turn a century-old railway into a nine-kilometre AI city prototyping ground.

Railway = spine · east-west links = stitches · three grounds = industry-to-life loop

Author: Wang Haoran (王浩冉) · Conceptual urban design submission

By 2035 Jing–Zhang is not a technology showroom, but a civic intelligence spine that people can enter, industries can test in and the city can learn from.

**RAIL**

civic intelligence spine

WINGS

transfer + applications

GROUND

verify + co–create + live

OS

rights + evidence + stop

Contents + reading route

Spatial idea first; industry and mobility next; three detailed grounds, twelve scenarios and delivery last.

01	01 Define the proposition	01—04	02	02 Build the spatial skeleton	05—14
03	03 Make industry operate	15—20	04	04 Design three open grounds	21—32
05	05 Deliver daily AI life	33—40			

SPATIAL JUDGEMENT

Spatial idea first; industry and mobility next; three detailed grounds, twelve scenarios and delivery last.

DESIGN ACTIONS

Five chapters answer why, where, how industry operates, how three key areas are designed, and how accountable AI becomes daily urban life.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Task–response matrix”.

Task–response matrix

Each of the six official tasks is answered by space, figure, metric and evidence.

01 Development vision → masterplan figure + metric + source	02 AI ecosystem → industry chain figure + metric + source
03 Urban form → three detailed plans figure + metric + source	04 Mobility + services → five stitches figure + metric + source
05 Infrastructure → component kit figure + metric + source	06 Heritage park → cultural commons figure + metric + source

SPATIAL JUDGEMENT

Each of the six official tasks is answered by space, figure, metric and evidence.

DESIGN ACTIONS

Every brief item needs a design claim, spatial figure, verifiable metric and source. Without all four, coverage is incomplete.

WHY THE NEXT PAGE

With coverage established, the proposal now moves from the brief to the real corridor.

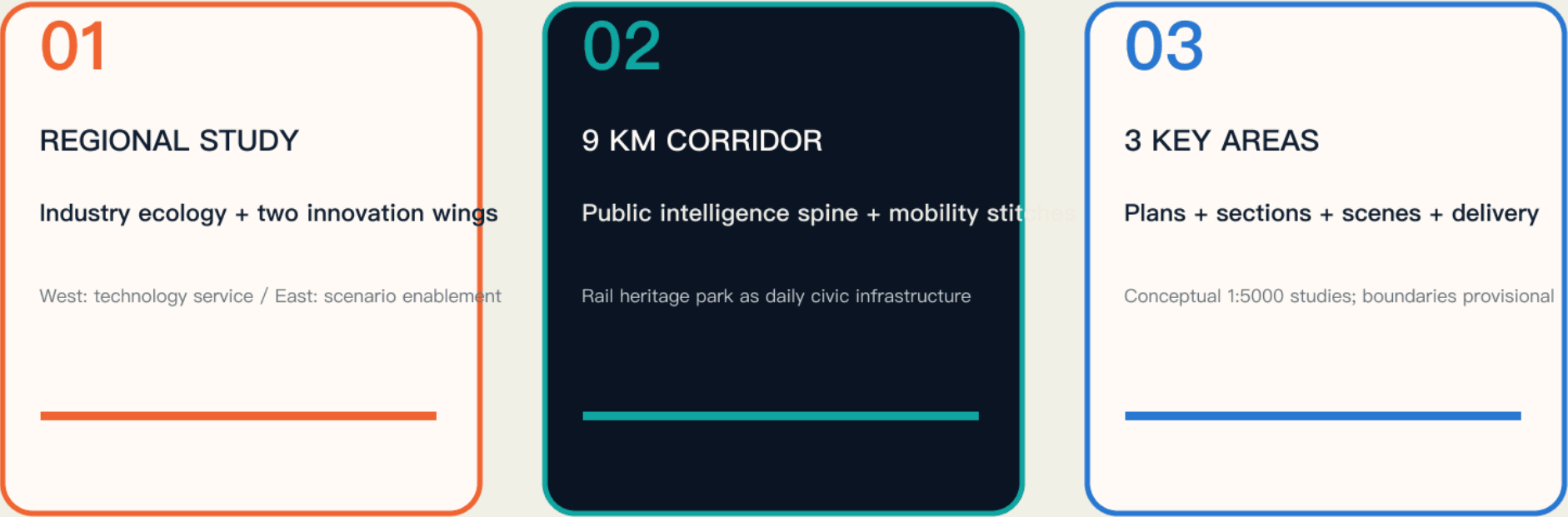
Three design scales

Regional study, nine–kilometre urban design and three detailed key areas become progressively specific.

02 / SCOPE

Three scales, one design argument

Research → overall design → detailed prototypes



The proposal gets more spatially specific as evidence becomes more local.

SPATIAL JUDGEMENT

Regional study, nine–kilometre urban design and three detailed key areas become progressively specific.

DESIGN ACTIONS

The study area connects resources; the nine–kilometre area establishes a shared skeleton; three key areas resolve plans, sections and scenes.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Existing assets”.

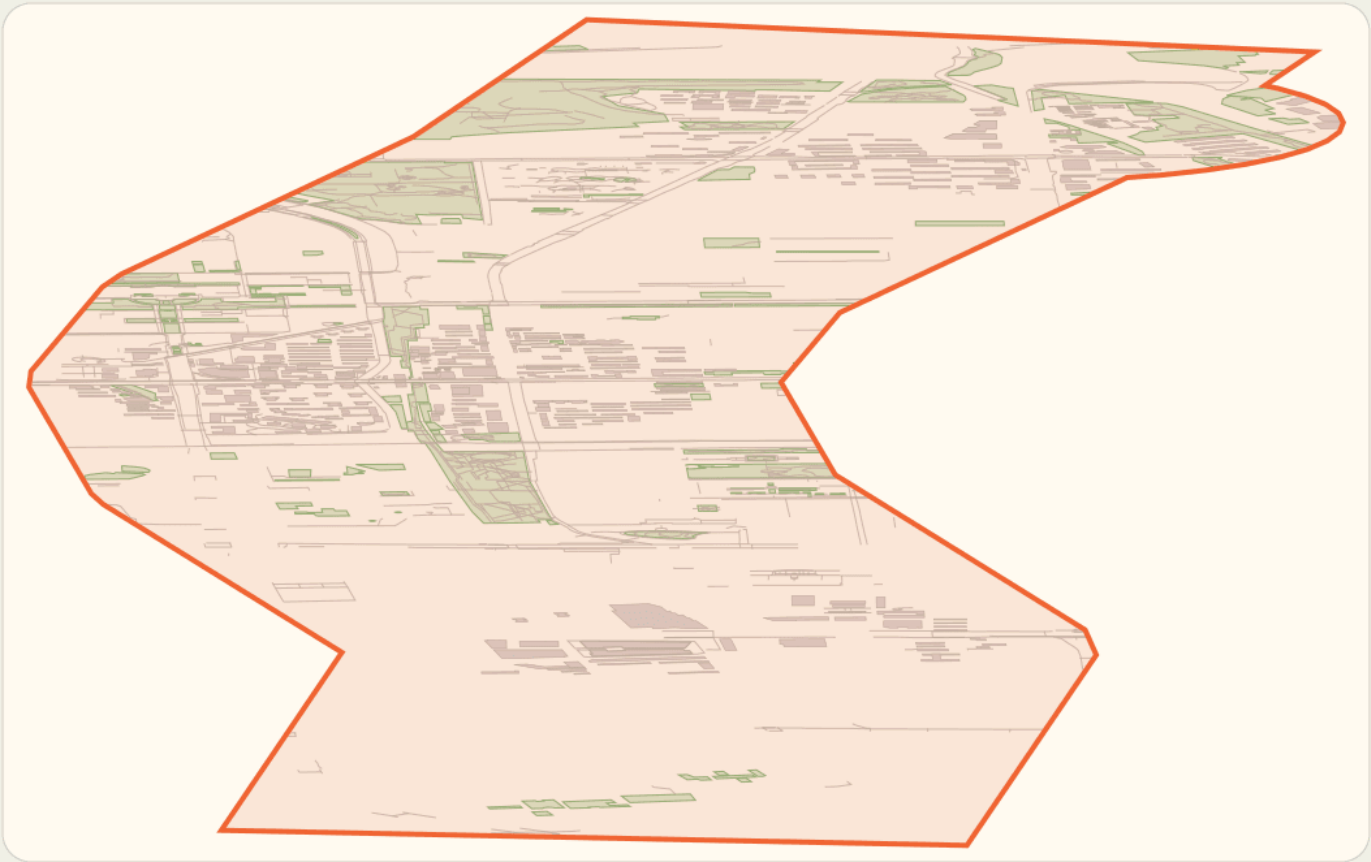
Existing assets

Rail heritage, park, universities, industry and neighbourhoods form one innovation infrastructure.

EXISTING ASSETS

Innovation already exists along the railway

Heritage · park · universities · industry · communities · metro



- 01

RAIL HERITAGE
- 02

PUBLIC PARK
- 03

UNIVERSITIES
- 04

AI INDUSTRY
- 05

NEIGHBOURHOODS
- 06

METRO HUBS

Design connects assets; it does not erase their existing

SPATIAL JUDGEMENT

Rail heritage, park, universities, industry and neighbourhoods form one innovation infrastructure.

DESIGN ACTIONS

Rail heritage and mature parks are retained, then linked to universities, parks, stations and neighbourhoods. The proposal starts from assets, not an isolated enclave.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Three breaks”.

Three breaks

Industry transfer, east–west mobility and everyday–life gaps must be repaired together.

1

INDUSTRY

Research reaches application too slowly



2

MOBILITY

Rail corridor still interrupts east–west walking



3

LIFE

AI remains an event, not an everyday service



SPATIAL JUDGEMENT

Industry transfer, east–west mobility and everyday–life gaps must be repaired together.

DESIGN ACTIONS

Industry lacks a validation path, walking lacks east–west continuity and residents lack daily AI services. One spatial strategy repairs all three.

WHY THE NEXT PAGE

This decision creates the need for the next page: “What is Open Ground?”.

What is Open Ground?

Railway pioneering spirit, open civic space and open–source innovation become one spatial idea.

SPATIAL IDEA

Open Ground is a city system, not a slogan

Pioneering railway spirit x open civic space x open–source innovation



The line becomes a public intelligence spine; the crossings become open grounds.

SPATIAL JUDGEMENT

Railway pioneering spirit, open civic space and open–source innovation become one spatial idea.

DESIGN ACTIONS

Pioneering recalls the railway, openness enables civic access and open source enables collaboration. Together they define construction and operation.

WHY THE NEXT PAGE

The proposition is now defined; the next page draws it as one masterplan.

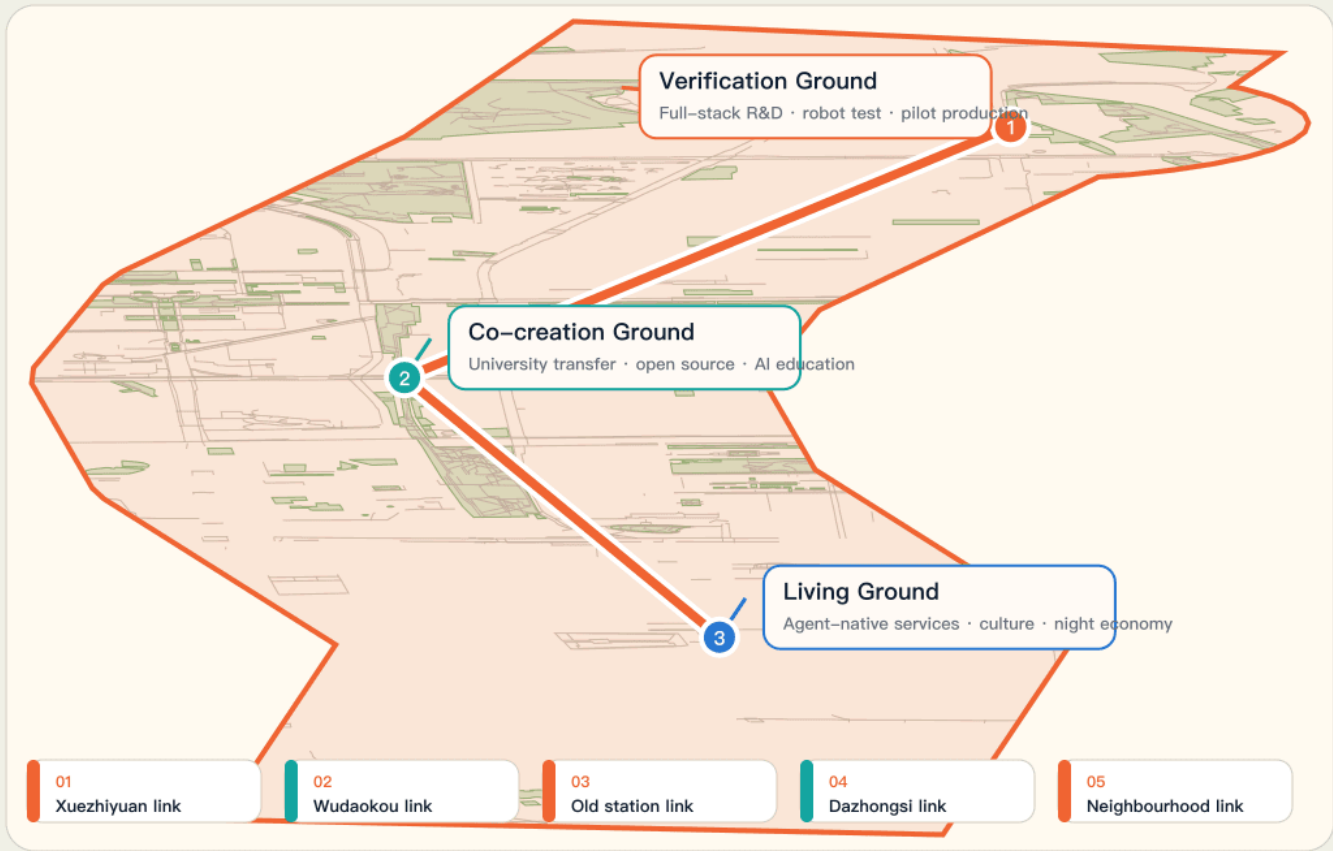
Nine-kilometre masterplan

One public intelligence spine, two innovation wings, three open grounds and five stitches.

01 / MASTERPLAN

A nine-kilometre public intelligence spine

One spine · two wings · three open grounds · five cross-corridor stitches



THREE OPEN GROUNDS

- 1

Verification Ground
Full-stack R&D · robot test · pilot production
- 2

Co-creation Ground
University transfer · open source · AI education
- 3

Living Ground
Agent-native services · culture · night economy

FACT / MODEL / DESIGN
Official facts remain distinct from provisional geometry.

SPATIAL JUDGEMENT

One public intelligence spine, two innovation wings, three open grounds and five stitches.

DESIGN ACTIONS

The heritage park is the civic intelligence spine; the west wing provides transfer and global services; the east wing enables embodied AI and urban applications; three grounds close the loop.

WHY THE NEXT PAGE

The masterplan establishes the skeleton; the next five pages explain its operating systems.

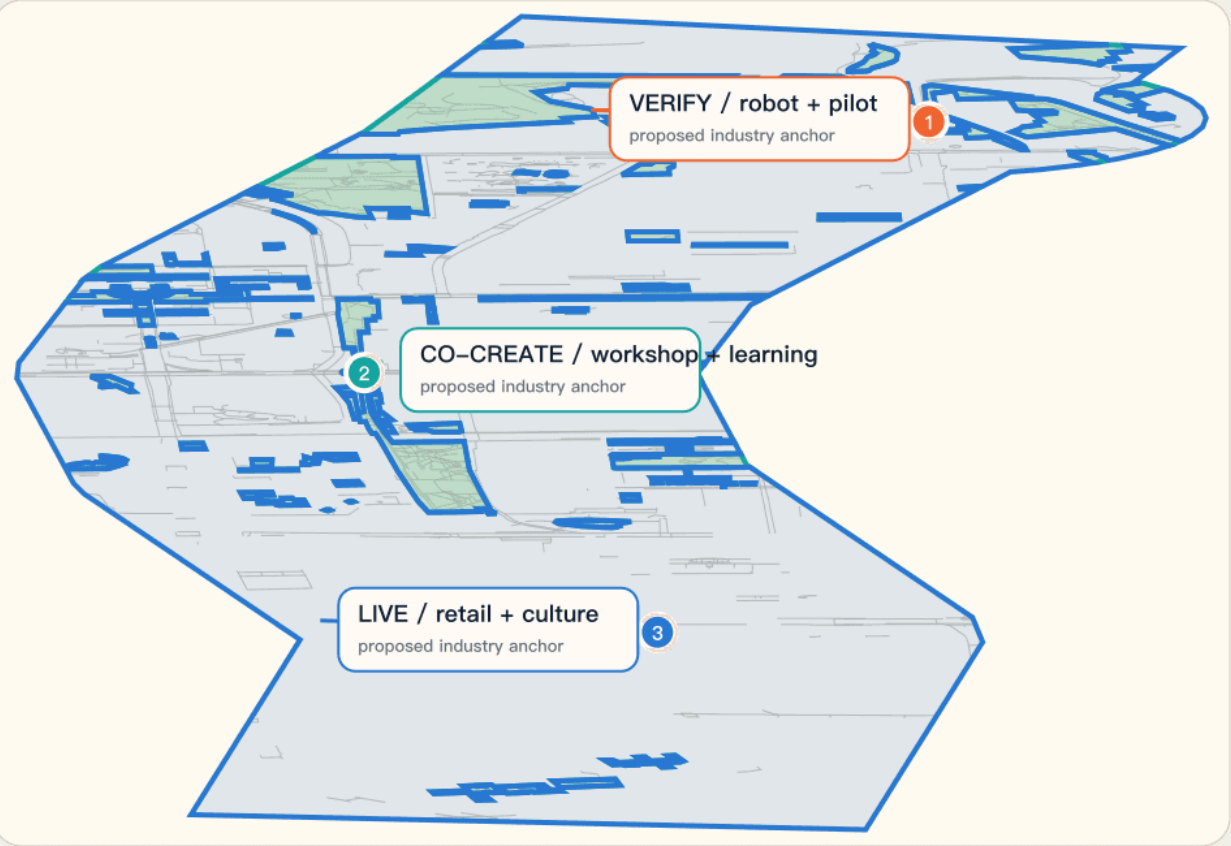
Overall programme

The corridor supports public learning and experience; two wings provide professional services and applications.

03 / LAND + INDUSTRY

Production spaces are arranged as a gradient

Lab → open workshop → validation → pilot production → city application



Existing land-use facts are not overwritten. Coloured layers are con

SPATIAL JUDGEMENT

The corridor supports public learning and experience; two wings provide professional services and applications.

DESIGN ACTIONS

Public learning, display and walking occupy the spine; professional R&D and applications occupy the wings; five stitches prevent simple linear accumulation.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Land + industry”.

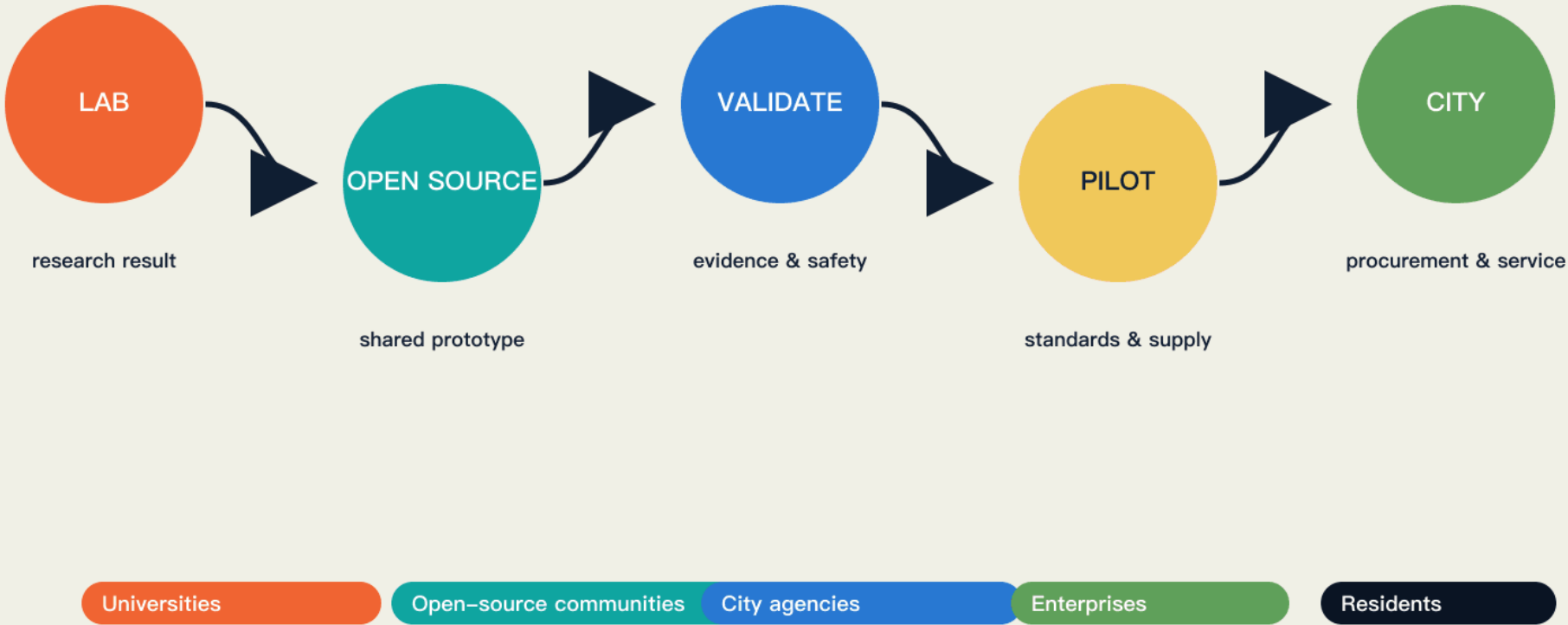
Land + industry

Research, open source, validation, pilot production and application form a continuous spatial gradient.

06 / INDUSTRY CHAIN

From research result to city service in five spatial moves

Every stage has a place, an operator and an exit condition



SPATIAL JUDGEMENT

Research, open source, validation, pilot production and application form a continuous spatial gradient.

DESIGN ACTIONS

Labs, workshops, validation gardens, pilot boxes and city interfaces are ordered by maturity, with feedback loops among the three grounds.

WHY THE NEXT PAGE

This decision creates the need for the next page: “East–west stitches”.

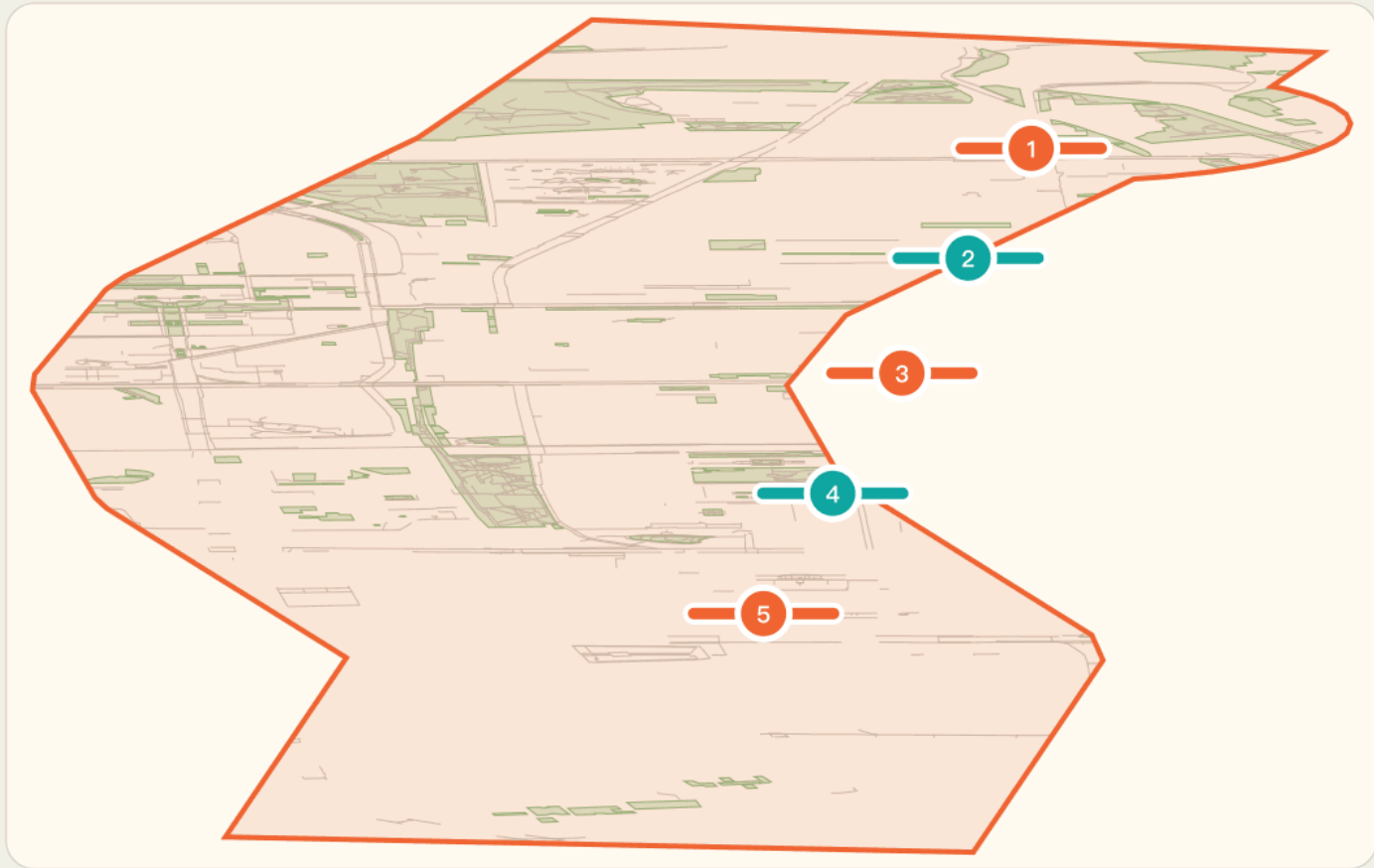
East–west stitches

500 m walking networks and 15–minute services identify priority interfaces.

04 / MOBILITY + BLUE-GREEN

Five stitches turn the corridor into a connector

500 m walking network · stations · crossings · 15–minute services



- 1 Xuezhuyuan—Zhongzhi
metro to validation garden
- 2 Wudaokou campus—street
courses + workshops + crossings
- 3 Qinghuayuan heritage station
heritage + question desk + rest
- 4 Dazhongsi four quadrants
wayfinding + accessible crossings
- 5 Neighbourhood—greenway
daily services + delivery hand-off

Priority: safe ground–level continuity before la

SPATIAL JUDGEMENT

500 m walking networks and 15–minute services identify priority interfaces.

DESIGN ACTIONS

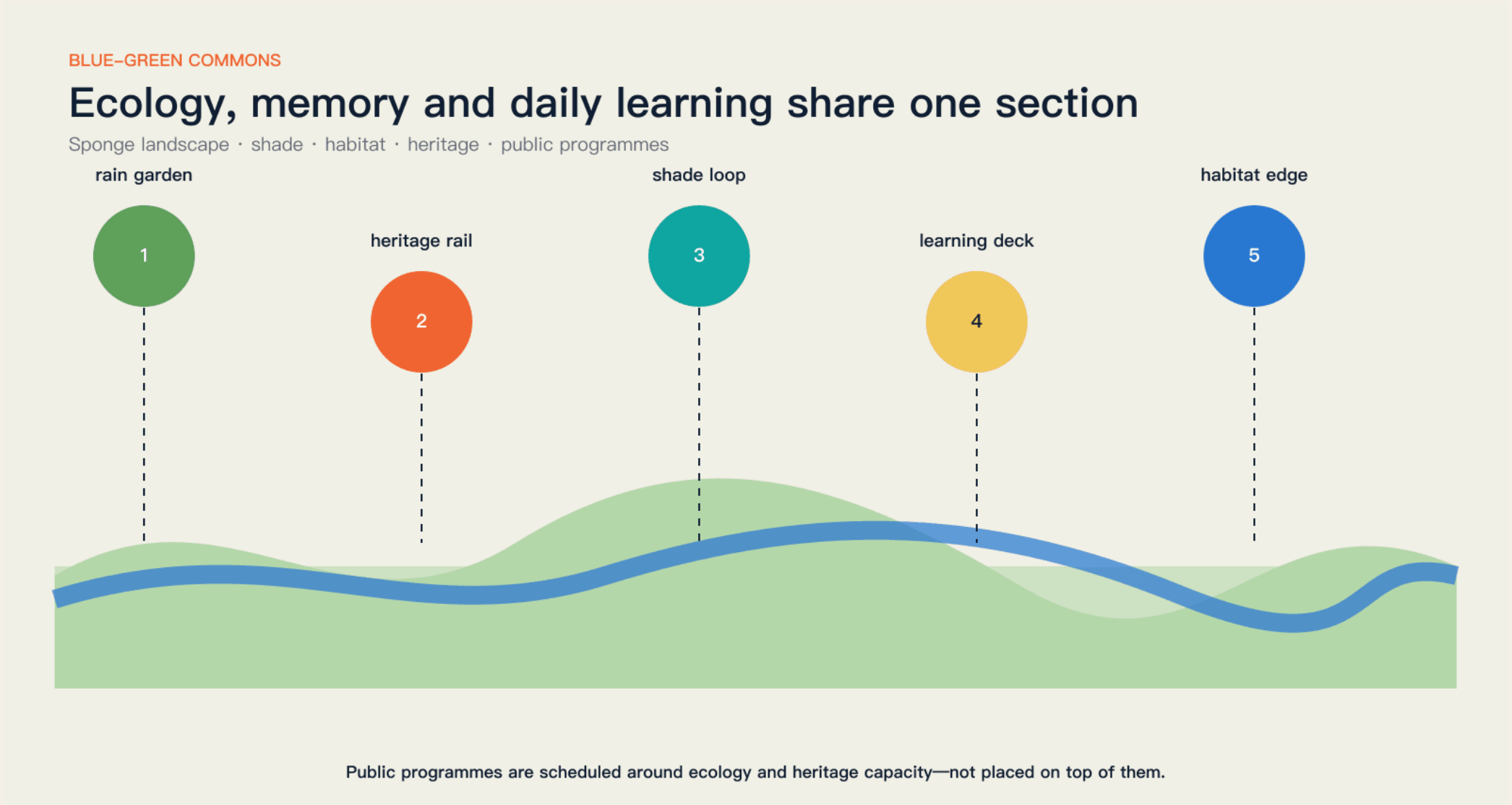
Five stitches connect metro, campus, heritage, four quadrants and neighbourhoods. Ground–level guidance, access and crossings come before major works.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Blue–green commons”.

Blue-green commons

The heritage park is both ecological corridor and everyday learning commons.



SPATIAL JUDGEMENT

The heritage park is both ecological corridor and everyday learning commons.

DESIGN ACTIONS

Shade, rain gardens, stopping places and public learning share one park section, so ecology and everyday use reinforce one another.

WHY THE NEXT PAGE

This decision creates the need for the next page: “New infrastructure”.

New infrastructure

Compute, data, robots, energy and sensing are embedded in maintainable small components.

05 / CIVIC AI INFRASTRUCTURE

Five small systems share one nine-kilometre service spine

Locate by task, maintain in public and expand only after evidence



- 1

Edge compute
three grounds / low-latency tests
- 2

Public data trust
AI Origin / governed access
- 3

Robot curb
verification + delivery interfaces
- 4

Energy microgrid
park service rooms + roofs
- 5

Urban sensing
five stitches + public dashboards
- BUILD SMALL · SHARE INTERFACES · PUBLISH STATUS

SPATIAL JUDGEMENT

Compute, data, robots, energy and sensing are embedded in maintainable small components.

DESIGN ACTIONS

Edge compute sits in three grounds; the data trust at AI Origin; robot curbs at test and delivery points; microgrids and sensing follow the civic spine.

WHY THE NEXT PAGE

With the civic infrastructure in place, the next chapter explains how industry moves through it.

Precedent translation

Living labs, city science, Fab Labs and participation platforms offer methods, not copied forms.

01

MIT City Science

Method translated: public interface, evidence or operation.

02

Amsterdam Living Lab

Method translated: public interface, evidence or operation.

03

Decidim

Method translated: public interface, evidence or operation.

04

Ars Electronica

Method translated: public interface, evidence or operation.

05

Paris SmartCity LL

Method translated: public interface, evidence or operation.

06

Fab Lab Barcelona

Method translated: public interface, evidence or operation.

SPATIAL JUDGEMENT

Living labs, city science, Fab Labs and participation platforms offer methods, not copied forms.

DESIGN ACTIONS

Precedents contribute methods—open tasks, visible tests, co-creation, long-term operation and public evidence—not copied urban form.

WHY THE NEXT PAGE

This decision creates the need for the next page: “AI industry chain”.

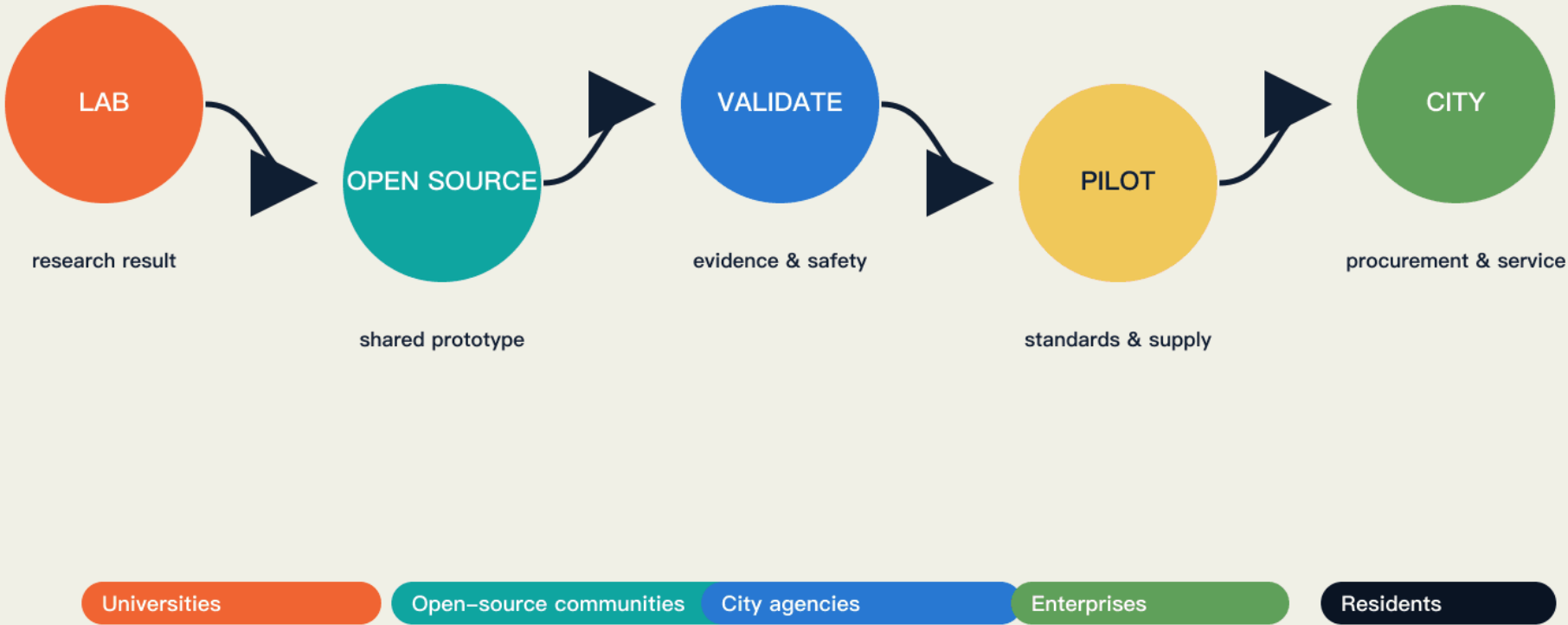
AI industry chain

Research outcomes enter the city through five operable spatial moves.

06 / INDUSTRY CHAIN

From research result to city service in five spatial moves

Every stage has a place, an operator and an exit condition



SPATIAL JUDGEMENT

Research outcomes enter the city through five operable spatial moves.

DESIGN ACTIONS

Each step has a place, operator and exit: universities deliver results, workshops prototype, test grounds produce evidence, pilots organise supply and cities procure services.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Three grounds + two wings”.

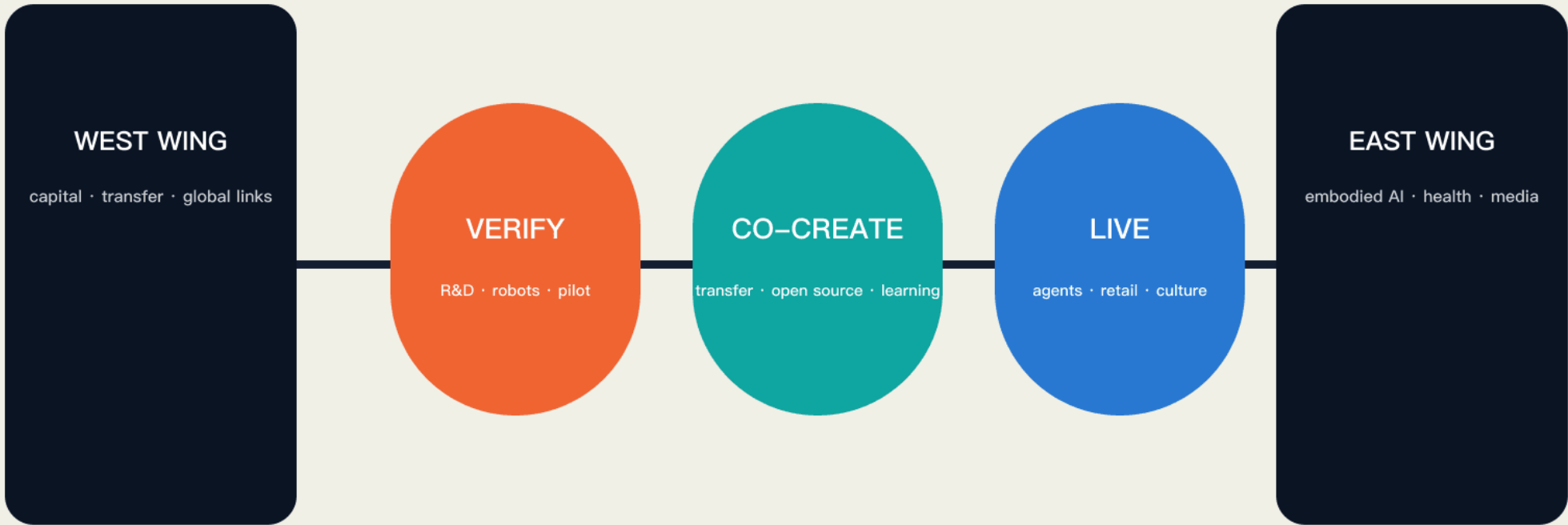
Three grounds + two wings

Zhongzhiyuan verifies, AI Origin co-creates, Dazhongsi brings AI into daily life.

THREE GROUNDS + TWO WINGS

Specialisation creates a complete innovation belt

Professional services to the west; applied scenarios to the east



One shared talent, data, evidence and maintenance network.

SPATIAL JUDGEMENT

Zhongzhiyuan verifies, AI Origin co-creates, Dazhongsi brings AI into daily life.

DESIGN ACTIONS

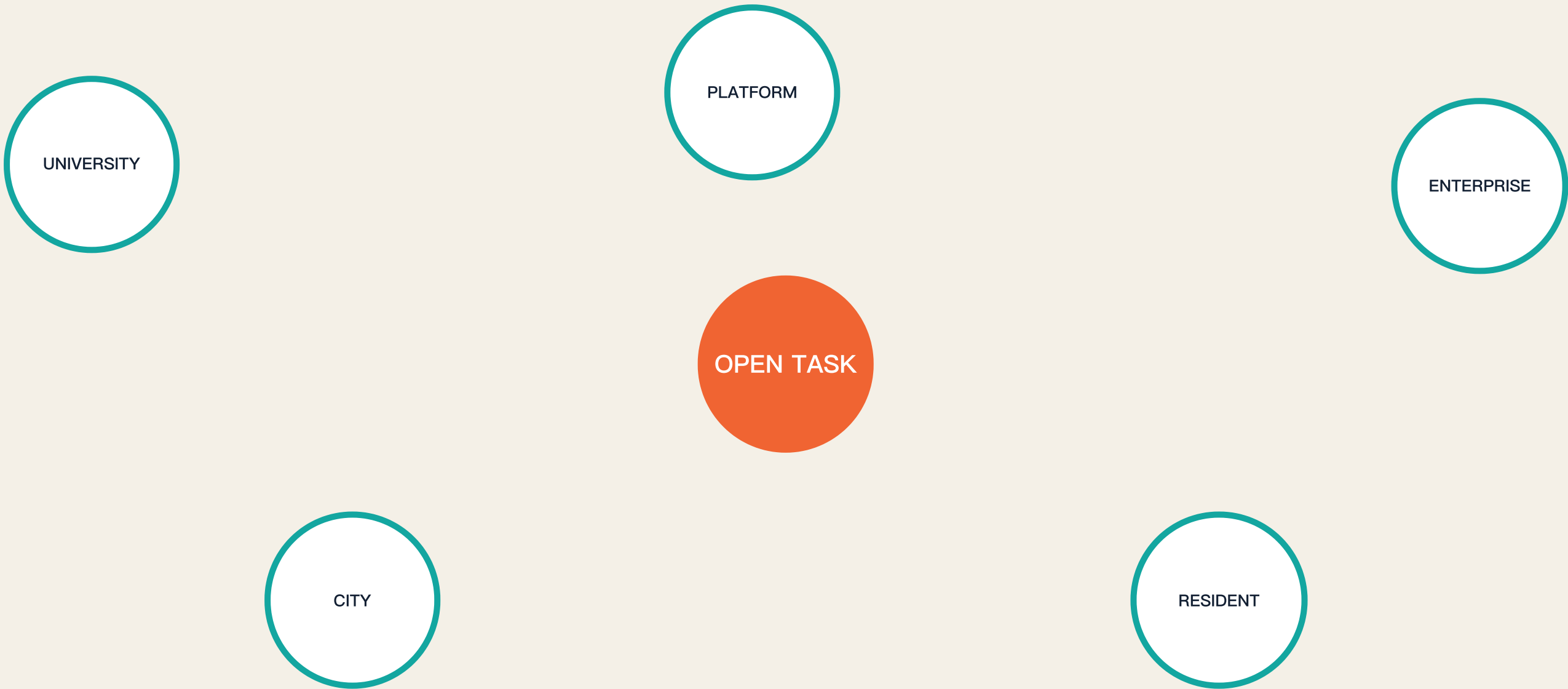
Zhongzhiyuan asks whether it can run safely; AI Origin whether it can be co-created; Dazhongsi whether it works in daily life. Two wings fill service gaps.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Innovation network”.

Innovation network

Universities, platforms, companies, agencies and residents connect around tasks, not park borders.



SPATIAL JUDGEMENT

Universities, platforms, companies, agencies and residents connect around tasks, not park borders.

DESIGN ACTIONS

Universities, platforms, firms, agencies and residents assemble around open tasks. Resources follow problems rather than park boundaries.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Five space prototypes”.

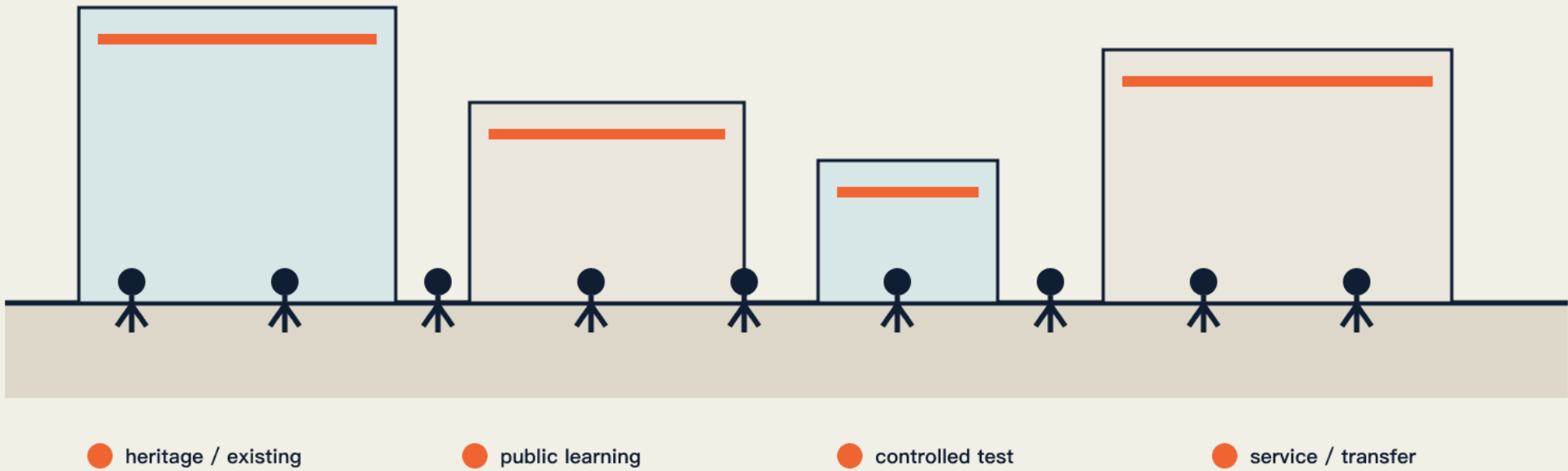
Five space prototypes

Lab, workshop, validation garden, pilot box and city interface combine as a kit of parts.

SECTION / PROTOTYPE

Verification garden section

People first · machines at walking speed · reversible components



Illustrative section: dimensions and engineering systems require professional design.

SPATIAL JUDGEMENT

Lab, workshop, validation garden, pilot box and city interface combine as a kit of parts.

DESIGN ACTIONS

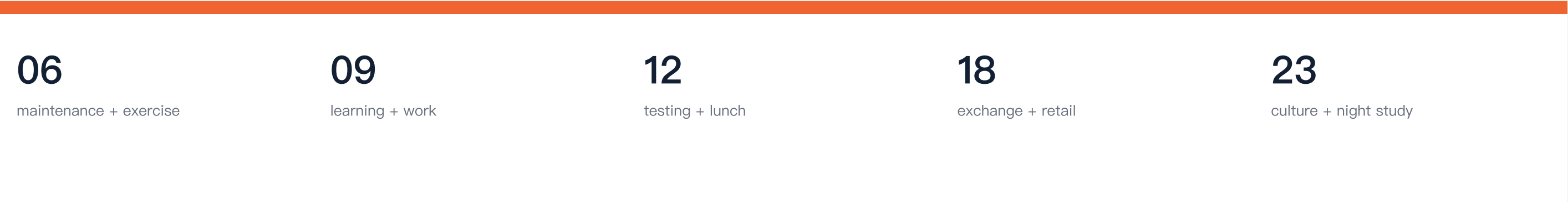
Five prototypes combine quiet research, open making, controlled testing, flexible pilots and public service. Ground-floor access determines real openness.

WHY THE NEXT PAGE

This decision creates the need for the next page: “24-hour talent chain”.

24-hour talent chain

Learning, work, testing, exchange, consumption and rest share space across the day.



SPATIAL JUDGEMENT

Learning, work, testing, exchange, consumption and rest share space across the day.

DESIGN ACTIONS

Morning maintenance, daytime learning, midday tests, evening exchange and night culture share one operating and safety framework.

WHY THE NEXT PAGE

Once the operating model is clear, the proposal enters the three key areas and draws them in detail.

Zhongzhiyuan plan

A validation garden links R&D, pilot production, export services and Xuezhiyuan station.



SPATIAL JUDGEMENT

A validation garden links R&D, pilot production, export services and Xuezhiyuan station.

DESIGN ACTIONS

A public entrance faces Xuezhiyuan station; R&D and pilots sit inside; a central garden hosts slow robots; export and display services line the edge.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Industry structure”.

Industry structure

Full-stack R&D inside, public demonstration outside, controlled tests within explicit limits.

01 R&D CORE

02 PILOT EDGE

03 PUBLIC DEMO

04 METRO LINK

SPATIAL JUDGEMENT

Full-stack R&D inside, public demonstration outside, controlled tests within explicit limits.

DESIGN ACTIONS

The R&D core stays quiet; pilots meet logistics; display faces the park; metro links serve talent and visitors.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Spatial section”.

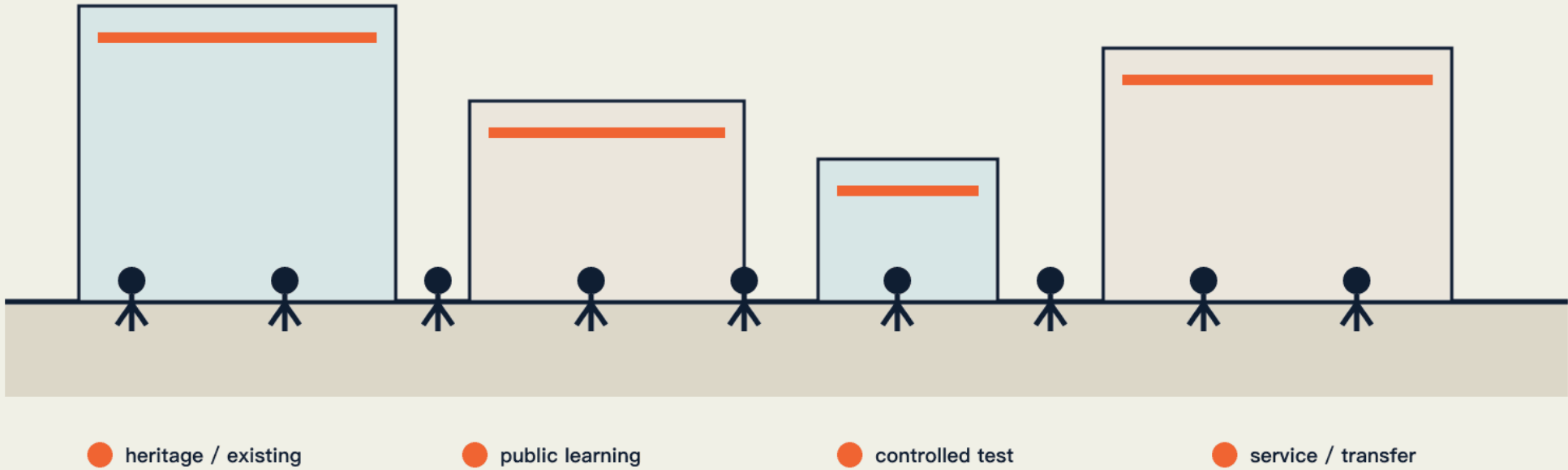
Spatial section

Machines move slowly; people have priority; observation and test zones remain connected but distinct.

SECTION / PROTOTYPE

Verification garden section

People first · machines at walking speed · reversible components



Illustrative section: dimensions and engineering systems require professional design.

SPATIAL JUDGEMENT

Machines move slowly; people have priority; observation and test zones remain connected but distinct.

DESIGN ACTIONS

The section separates walking, observation, stoppable test loops and production. Machines are speed-limited and people control crossings.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Controlled tests”.

Controlled tests

Robots and delivery systems begin in a stoppable and recoverable validation garden.



SPATIAL JUDGEMENT

Robots and delivery systems begin in a stoppable and recoverable validation garden.

DESIGN ACTIONS

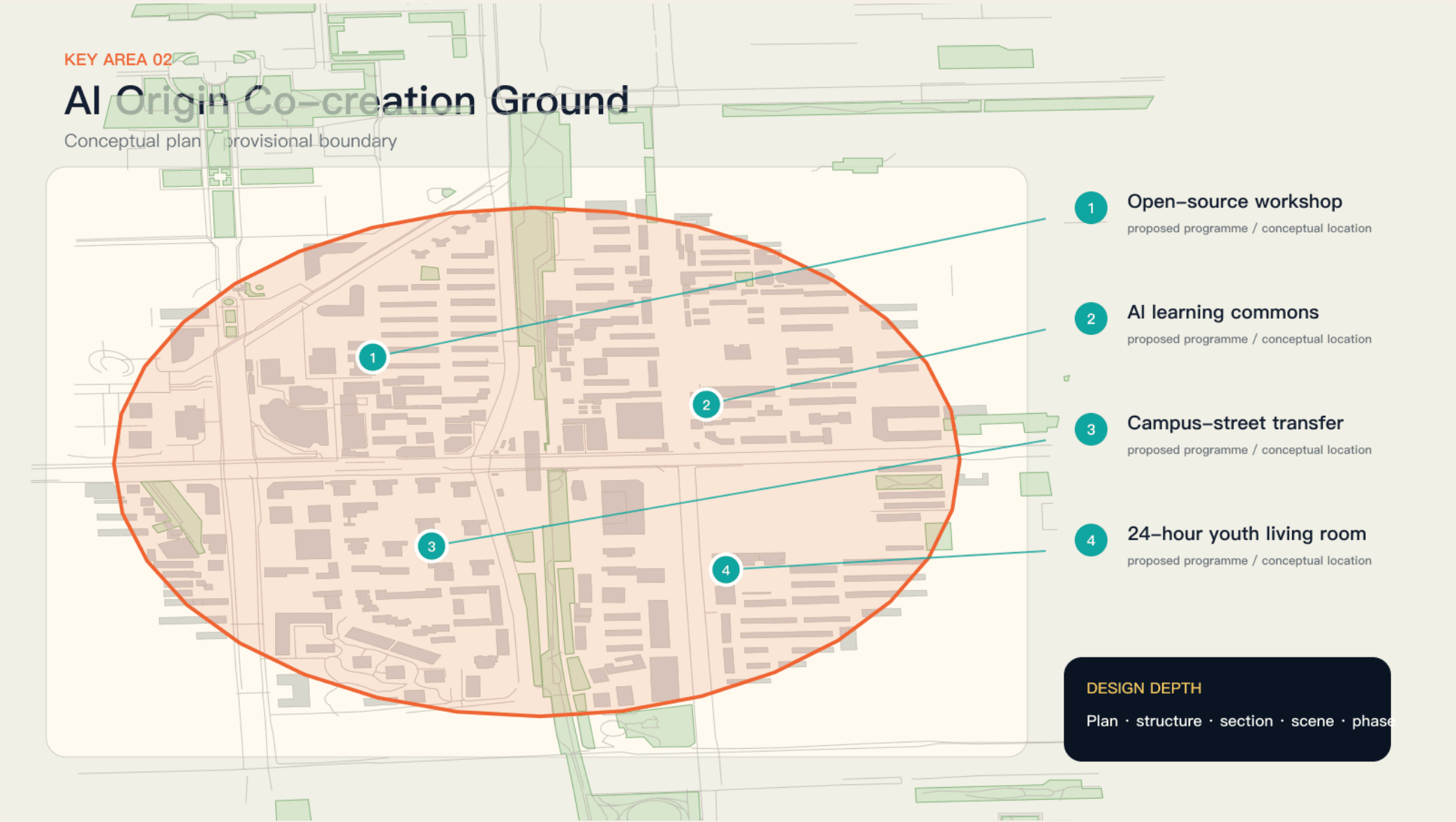
Tests start in fixed times and areas with public status, incident logs and stop controls, then expand to delivery hand-offs only after evidence.

WHY THE NEXT PAGE

The verification ground is complete; the next section moves from testing to university-led co-creation.

AI Origin plan

University transfer, open source, AI education and youth life form a one-kilometre learning circle.



SPATIAL JUDGEMENT

University transfer, open source, AI education and youth life form a one-kilometre learning circle.

DESIGN ACTIONS

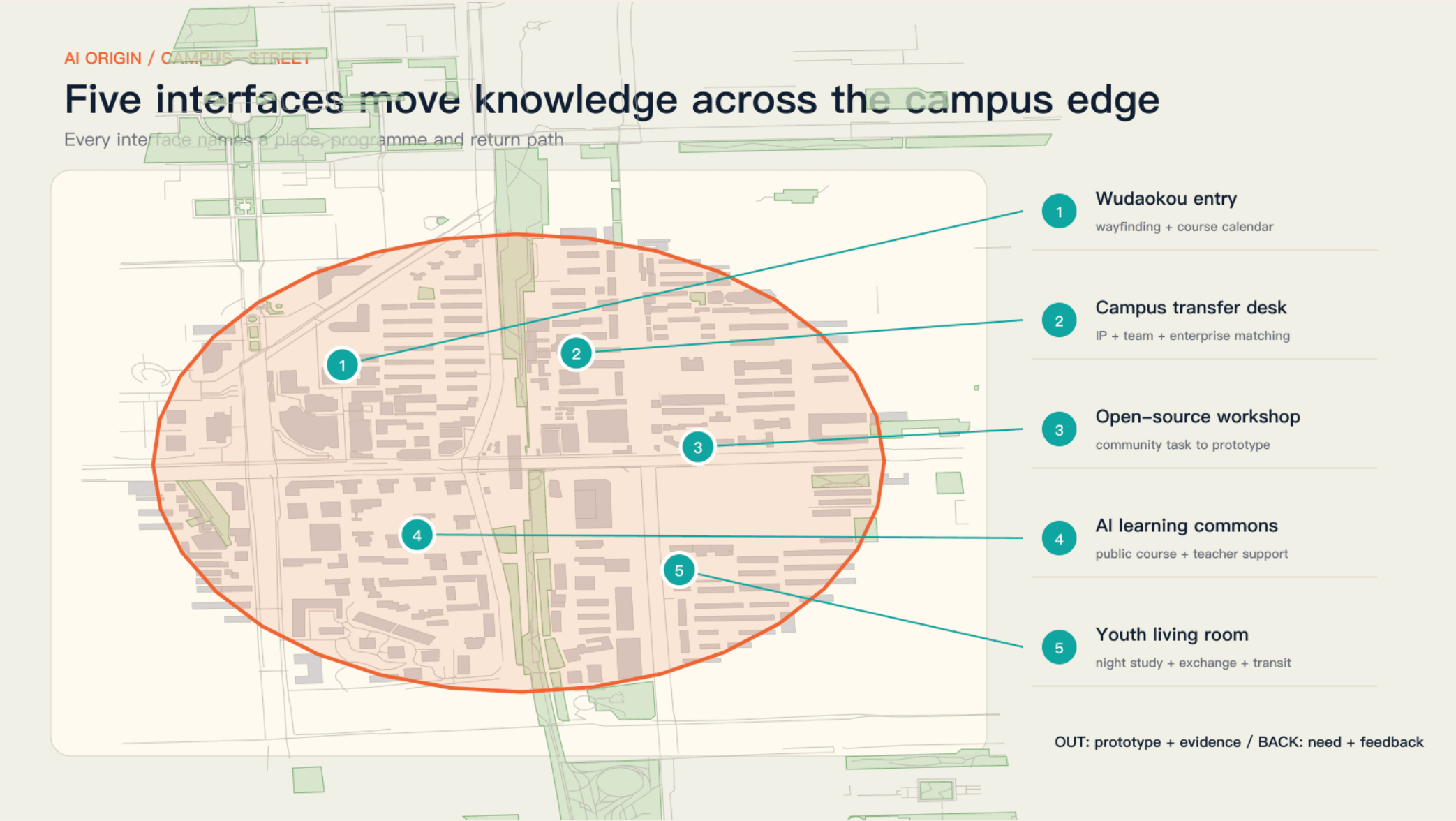
Wudaokou and campus edges become entries to a one-kilometre learning circle of transfer services, workshops, AI learning and youth life.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Campus-street network”.

Campus–street network

Wudaokou translates campus capability into public courses and workshops.



SPATIAL JUDGEMENT

Wudaokou translates campus capability into public courses and workshops.

DESIGN ACTIONS

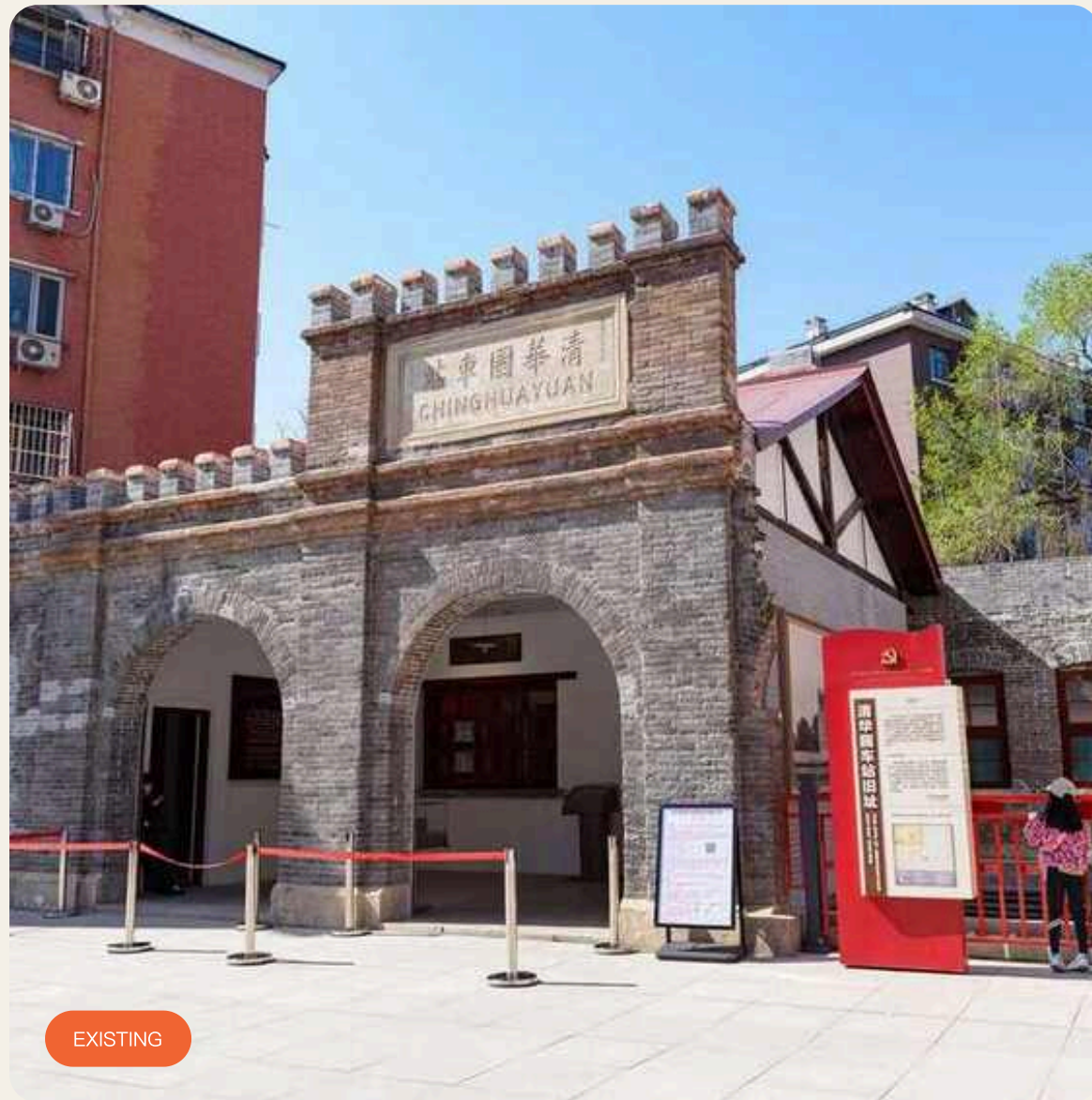
Courses, question calls and workshops move research into the street; needs and feedback return to research teams.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Open workshop + AI education”.

Open workshop + AI education

Real problems enter the workshop; prototypes and evidence return to the community.



EXISTING



CONCEPT

SPATIAL JUDGEMENT

Real problems enter the workshop; prototypes and evidence return to the community.

DESIGN ACTIONS

Workshops accept real community problems and return demonstrable prototypes, instructions and evidence. Learning and production share one place.

WHY THE NEXT PAGE

This decision creates the need for the next page: "Youth living room".

Youth living room

Twenty-four-hour innovation moves out of offices into public places for staying and exchange.



SPATIAL JUDGEMENT

Twenty-four-hour innovation moves out of offices into public places for staying and exchange.

DESIGN ACTIONS

Open ground floors combine study tables, pitches, food and night transport information, enabling people to stay rather than only commute.

WHY THE NEXT PAGE

Co-creation now has a spatial home; the next section brings AI into daily consumption and culture.

Dazhongsi plan

Four quadrants work together as an AI living open ground.



SPATIAL JUDGEMENT

Four quadrants work together as an AI living open ground.

DESIGN ACTIONS

Unified guidance and ground continuity connect four quadrants, then smart retail, culture, community service and interchange create an everyday open ground.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Four-quadrant stitch”.

Four-quadrant stitch

Unified ground guidance, accessible continuity and cycle crossings repair legibility first.



SPATIAL JUDGEMENT

Unified ground guidance, accessible continuity and cycle crossings repair legibility first.

DESIGN ACTIONS

Number quadrants and exits, add continuous ramps, cycle waiting and lighting, and name the public destinations in every direction.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Agent-native commerce”.

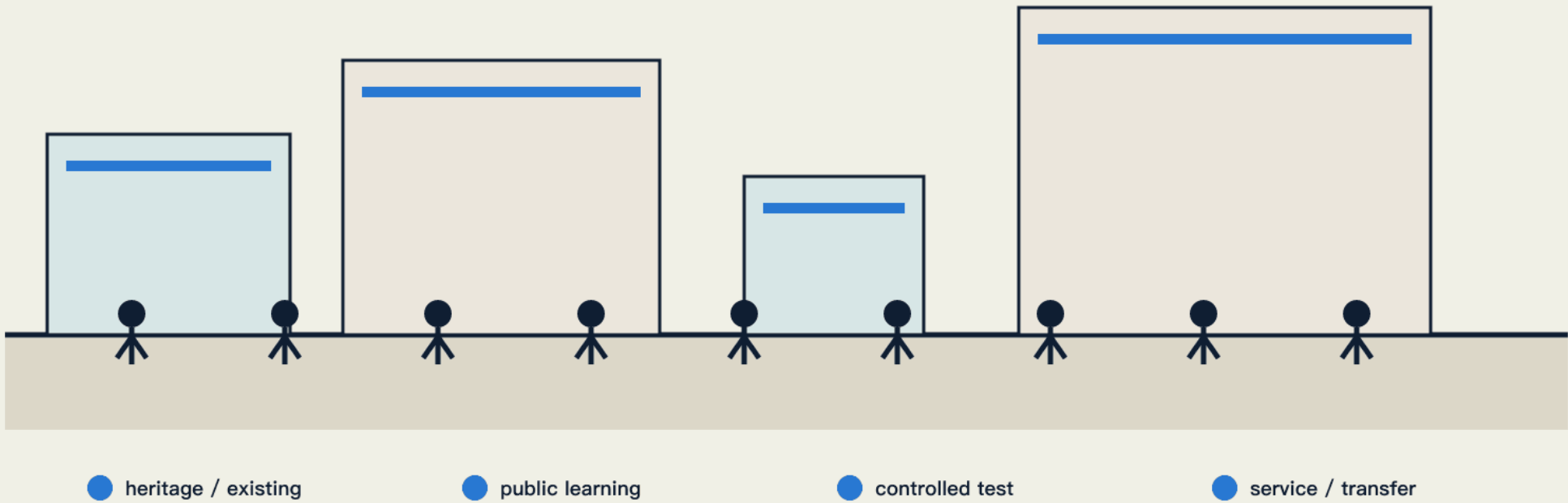
Agent-native commerce

Devices, content, community services and metro interchange share active ground floors.

SECTION / PROTOTYPE

Four-quadrant living section

People first · machines at walking speed · reversible components



Illustrative section: dimensions and engineering systems require professional design.

SPATIAL JUDGEMENT

Devices, content, community services and metro interchange share active ground floors.

DESIGN ACTIONS

The transit hall manages interchange; active ground floors show devices and content; upper levels hold community creation; back-of-house supports delivery and maintenance.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Culture + night co-stage”.

Culture + night co-stage

Night economy and railway memory coexist without turning heritage into a technology prop.



SPATIAL JUDGEMENT

Night economy and railway memory coexist without turning heritage into a technology prop.

DESIGN ACTIONS

Rail memory enters night programmes through sound, image and storytelling. Commercial activity reserves time and space for rest, access and community content.

WHY THE NEXT PAGE

Three districts are resolved; the final chapter connects them through daily scenarios and delivery.

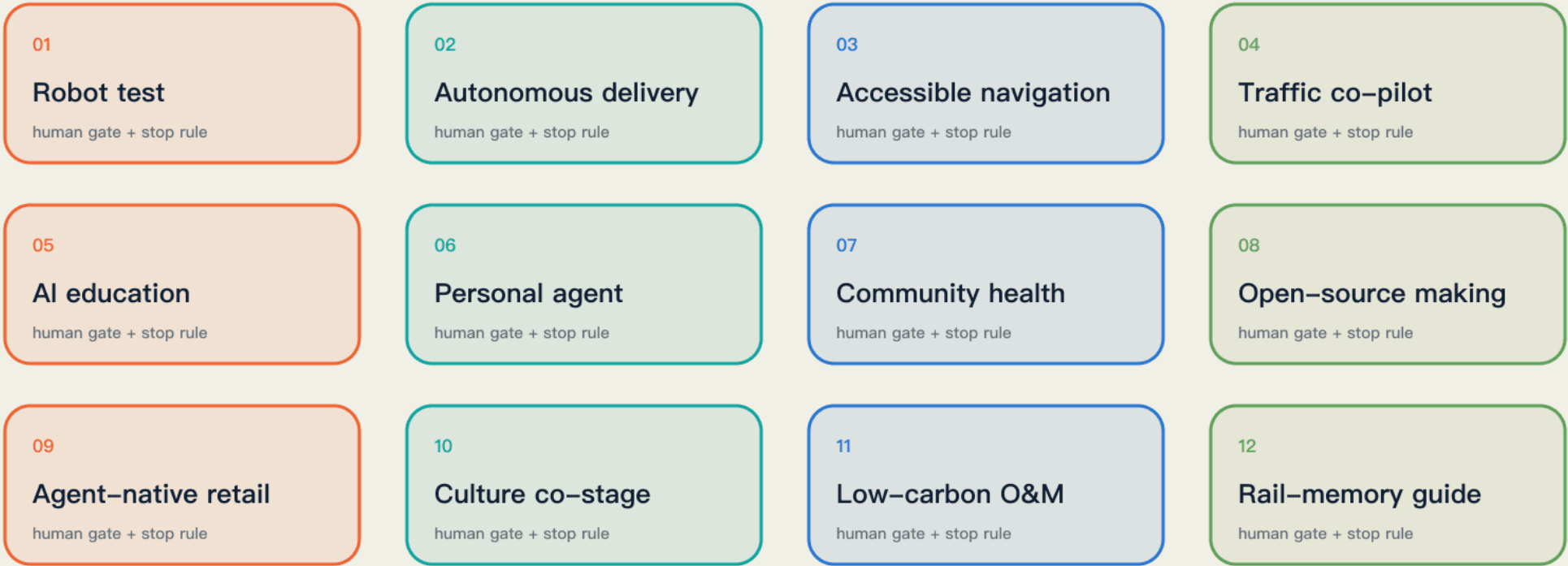
Twelve–scenario matrix

Mobility, health, education, robots, culture and operations share one accountability system.

07 / AI CITY LIFE

Twelve visible scenarios, one accountable operating system

The public experiences services; agents coordinate evidence, consent and stopping



SPATIAL JUDGEMENT

Mobility, health, education, robots, culture and operations share one accountability system.

DESIGN ACTIONS

Twelve front–stage experiences are distributed across three grounds but share rights, evidence, safety and archive systems. Each names users, place and stop rule.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Mobility + robots”.

Mobility + robots

Low-speed, visible and stoppable rules come before scaling.



EXISTING



CONCEPT

SPATIAL JUDGEMENT

Low-speed, visible and stoppable rules come before scaling.

DESIGN ACTIONS

Robots, autonomous shuttles and delivery start on slow, visible and supervised routes. Kerb hand-offs keep machines out of continuous walking space.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Health, learning + inclusion”.

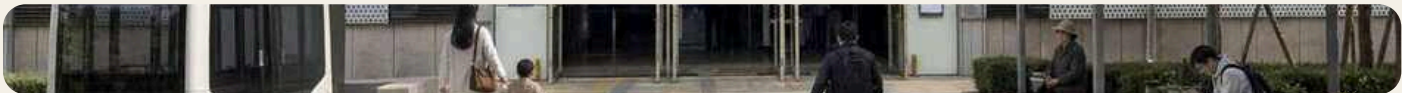
Health, learning + inclusion

Technology first improves daily access for older people, children, disabled people and carers.

culture



inclusive mobility



public question desk



SPATIAL JUDGEMENT

Technology first improves daily access for older people, children, disabled people and carers.

DESIGN ACTIONS

Community health does not replace doctors and AI learning does not replace teachers. Accessible navigation prioritises older, young, disabled and caring users.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Rail culture + kit of parts”.

Rail culture + kit of parts

Wayfinding, question desks, displays, seating and micro–utility rooms use maintainable modules.

RAIL CULTURE + KIT OF PARTS

A recognisable family of small, maintainable interfaces

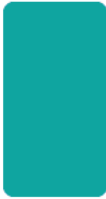
No heritage decoration; each component carries a real urban function



01

WAYFINDING

reversible · repairable



02

QUESTION DESK

reversible · repairable



03

DISPLAY RAIL

reversible · repairable



04

REST + SHADE

reversible · repairable



05

MICRO UTILITY

reversible · repairable



06

HONOUR WALL

reversible · repairable

SPATIAL JUDGEMENT

Wayfinding, question desks, displays, seating and micro–utility rooms use maintainable modules.

DESIGN ACTIONS

Wayfinding, question desks, display rails, shade, micro–utilities and honour walls share a material system and can be replaced one unit at a time.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Agent operating system”.

Agent operating system

Six agents coordinate; consent, professional decision, public review and stop/restore remain human gates.

08 / OPERATING SYSTEM

Six agents coordinate; four human gates decide

Governance is compressed into one page and supports every spatial scenario



SPATIAL JUDGEMENT

Six agents coordinate; consent, professional decision, public review and stop/restore remain human gates.

DESIGN ACTIONS

Six agents organise questions, evidence, matching, safety, rights and archives. Consent, expert judgement, public review and stop/restore remain human decisions.

WHY THE NEXT PAGE

This decision creates the need for the next page: “90 days—1 year—3 years”.

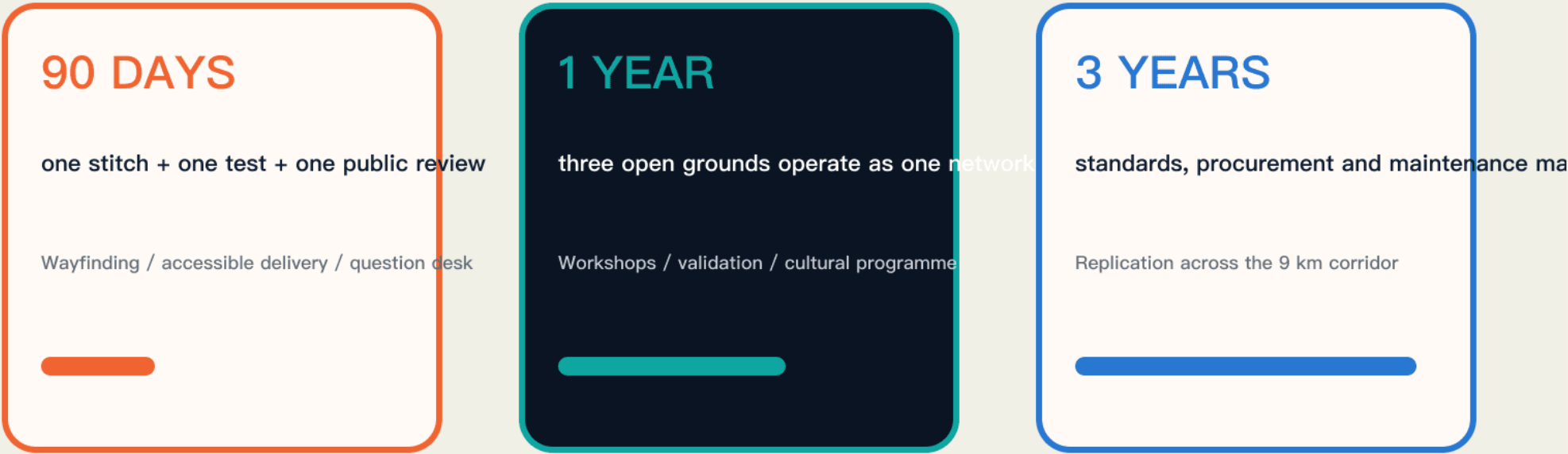
90 days—1 year—3 years

Start with one stitch and one test, then connect three grounds and replicate along nine kilometres.

09 / DELIVERY

Prove a small loop, then scale the network

90 days → 1 year → 3 years



Advance only when evidence, maintenance ownership and stopping conditions are explicit.

SPATIAL JUDGEMENT

Start with one stitch and one test, then connect three grounds and replicate along nine kilometres.

DESIGN ACTIONS

In 90 days build one reversible stitch and test; in one year connect three grounds; in three years replicate only components supported by evidence.

WHY THE NEXT PAGE

This decision creates the need for the next page: “Metrics, evidence + risk”.

Metrics, evidence + risk

Every number returns to geometry; every concept has a status; risk is managed in one place.

10 / METRICS + EVIDENCE

One number, one geometry, one status

Official facts · open data · field observation · conceptual model · professional follow-up

13.333

km² provisional study model

recomputed from GeoJSON

8.82%

mapped green ratio

recomputed from GeoJSON

0.063%

candidate public-space ratio

recomputed from GeoJSON

117.0

km mapped walk/cycle network

recomputed from GeoJSON

OFFICIAL FACT

OPEN DATA

FIELD OBSERVATION

CONCEPT DESIGN

TO BE VERIFIED

The announced 11.4 km² task area remains a separate brief fac

SPATIAL JUDGEMENT

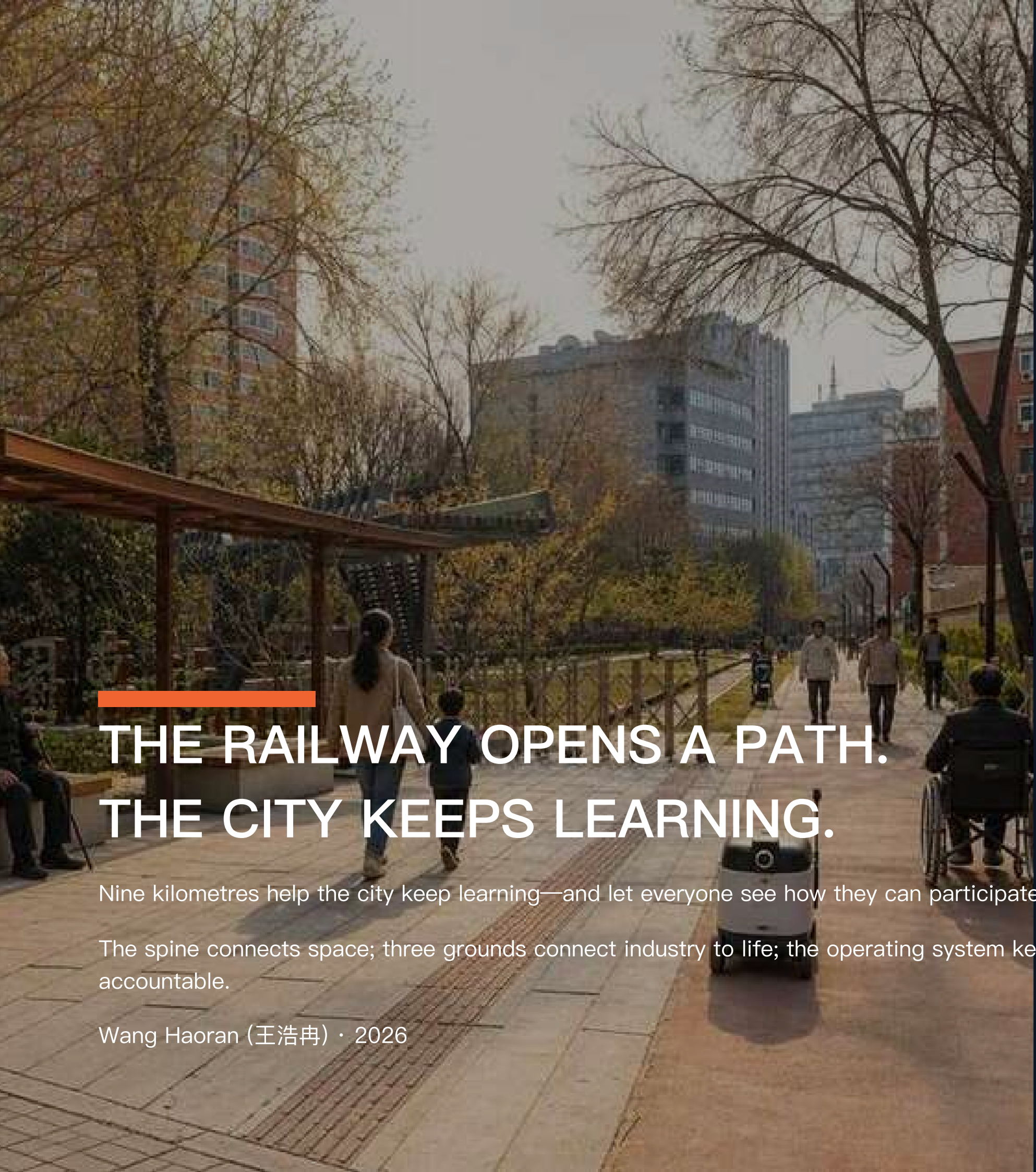
Every number returns to geometry; every concept has a status; risk is managed in one place.

DESIGN ACTIONS

Areas and ratios are recomputed from geometry; facts, observations, open data, design and unknowns remain distinct; risk is concentrated without obscuring design.

WHY THE NEXT PAGE

With metrics and risks made explicit, the final page returns to the complete future city image.



THE RAILWAY OPENS A PATH. THE CITY KEEPS LEARNING.

Nine kilometres help the city keep learning—and let everyone see how they can participate in its future.

The spine connects space; three grounds connect industry to life; the operating system keeps every test accountable.

Wang Haoran (王浩冉) · 2026

